

Module 7: Lesson 3

Reinforcement Learning: The Cliff Walk



Outline

- ▶ The cliff walk environment
- ▶ Performance of SARSA and Q-Learning in the cliff walk

The cliff walk environment

Before studying a fully-fledged financial application of RL, we are going to devote some additional analysis of simple implementations to SARSA and Q-learning.

Consider the gridworld shown below, with start and goal states, and the actions causing movements up, down, right, or left.

Rewards are -1 on all transitions except those into the region marked "THE CLIFF." If the agent steps into this region, it generates a reward of -100, and the agent returns to the start.

START	THE CLIFF										FINISH

The cliff walk environment: SARSA

The optimal policies from SARSA opt for a "safe" path that avoids the row of cells just above the cliff.

That is, choices of "up" (action 0) dominate above the cliff, while "right" dominates elsewhere and "down" once the agent has reached the right edge of the grid.

Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Down
Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Down
Up	Up	Up	Up	Up	Up	Up	Right	Up	Right	Right	Down
Up	THE CLIFF										FINISH

The cliff walk environment: Q-Learning

Q-learning follows a "riskier" path just one step away from the cliff. This results in the agent falling off the cliff because of the ϵ -greedy action selection.

That is, the "off-policy" method updates the state-action value based on the optimal action in the next step, the max operator, regardless of the actual uncertainty generated by the ϵ -greedy action selection.

Right	Right	Right	Right	Down	Right	Down	Right	Down	Right	Right	Down
Right	Right	Right	Right	Down	Right	Down	Right	Right	Down	Down	Down
Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Right	Down
Up	THE CLIFF										FINISH

By exploring the implementation of the cliff walk that we provide in the Jupyter Notebook for this lesson, you should be able to understand how both TD methods would yield, theoretically and/or numerically, the same outcome.

Summary of Lesson 3

In Lesson 3, we have looked at:

- The differences in performance between SARSA and Q-Learning in the cliff walk

⇒ **References for this lesson:**

Sutton, Richard S., and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018. (see Chapter 6)

TO DO NEXT: Now, please go to the associated Jupyter Notebook for this lesson to study the programming implementation of SARSA and Q-Learning in the cliff walk.

In the next lesson, we will develop a fully fledged financial implementation of Q-Learning in the context of optimal trading choices.